



AUTOORBIT: Physics-Informed Satellite Orbit Prediction

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Abstract

Accurate onboard orbit prediction is essential for autonomous satellite operations. Existing methods fall into two categories: physics-based and data-driven. However, both face critical limitations under realistic onboard conditions. Physics-based methods compute future satellite positions using physical models based on initial orbital states, but their performance degrades significantly onboard due to noisy input data and the lack of real-time model parameter updates. Data-driven methods avoid complex force modeling by learning directly from historical trajectory data, but they often violate fundamental physical laws, resulting in physically inconsistent trajectories. Moreover, both approaches suffer severe performance degradation during maneuvers. In this paper, we propose AUTOORBIT, a physics-informed orbit prediction framework for real-time onboard use. AUTOORBIT adopts a hierarchical decomposition of orbital behavior that enforces physical consistency across global orbital structure, local orbital dynamics, and discrete maneuver events. Specifically, it reconstructs a global reference orbit to prevent long-horizon drift, learns fine-grained dynamics with a physics-informed Fourier neural operator, and applies maneuver-aware variational correction to ensure continuity across scheduled maneuvers. We evaluate AUTOORBIT on three operational in-orbit

satellites, achieving 62.71 m position error at a 30-minute prediction horizon, over 60% improvement compared to baseline methods. We also evaluate system overhead by deploying AUTOORBIT on a flight-representative Jetson Orin Nano, demonstrating efficient resource utilization with stable power and thermal behavior.

CCS Concepts

• Applied computing → Aerospace; • Computing methodologies → Neural networks.

Keywords

Satellite orbit prediction; physics-informed; satellite computing

ACM Reference Format:

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1 Introduction

The rapid growth of Earth-orbiting satellites is fundamentally reshaping the near-Earth space environment. By 2030, more than 100,000 active satellites and over one million tracked debris objects are expected to coexist in low and medium Earth orbits [14]. This unprecedented congestion amplifies collision risks and places severe pressure on traditional, ground-centric satellite operations. In

response, the community is moving toward autonomous spacecraft that can independently sense, plan, and act without waiting for ground commands. At the core of this autonomy lies a fundamental scientific and operational problem: accurate, real-time prediction of satellite orbits onboard. High-precision orbit prediction is no longer only a classical astrodynamics task, but a critical enabler for a wide range of science and engineering objectives, including agile Earth observation planning [28, 33], fast inter-satellite communication alignment [29], and proactive collision avoidance [27].

Satellite orbit prediction is a classical problem in astrodynamics, governed by well-defined laws of orbital dynamics. Satellite motion is primarily driven by Earth's gravity, while being continuously influenced by perturbations, including atmospheric drag, third-body gravitation from the Moon and Sun, and occasional onboard maneuvers. The superposition of these forces results in highly complex orbital dynamics. In current practice, onboard orbit prediction predominantly relies on *physics-based analytical models*, most notably Simplified General Perturbations 4 (SGP4) [20, 36]. These models use closed-form approximations with simplified force representations and coarse orbital summaries (e.g., Two-Line Elements (TLEs)) derived from ground tracking data [53, 57]. Their low computational overhead makes them suitable for resource-constrained satellites, but the simplified modeling leads to kilometer-level position errors [51], which is insufficient for precision-critical autonomous tasks. In contrast, *physics-based numerical propagators*, such as High Precision Orbit Propagator (HPOP) [3, 10, 36], integrate orbital motion using high-precision physical models and detailed perturbation formulations. These methods represent the state of the art in orbit prediction accuracy and are widely used in ground-based applications, including ground station antenna tracking, orbit analysis, and long-term mission planning [3, 10].

Given the high accuracy of numerical orbit propagators, a natural question arises: can these methods be deployed onboard to support real-time autonomous operations? In practice, however, their performance degrades significantly in space due to two fundamental limitations: imprecise input data and the lack of real-time environmental parameter updates. First, numerical methods exhibit strong sensitivity to input accuracy, particularly the satellite's initial orbital state defined by position and velocity vectors. Even small errors can accumulate rapidly during numerical integration, leading to severe accuracy degradation over time [6, 30]. Onboard, initial states are typically estimated from Global Navigation Satellite System (GNSS) measurements [17], which contain non-negligible noise. Under realistic GNSS noise conditions, our empirical evaluation shows that 48-hour prediction errors can increase by up to $100\times$ (§2.2). Second, these methods depend on continuous updates of environmental parameters, typically streamed from global ground networks. However, many satellites lack persistent ground contact and cannot receive these updates in real time [12]. This limits the model's ability to account for time-varying forces and undermines its suitability for onboard deployment.

Beyond physics-based models, another approach applies data-driven methods to orbit prediction. These include end-to-end models that regress future orbital states from past sequences [32], and hybrid models that learn to correct outputs from analytical models like SGP4 [18, 42]. Data-driven methods learn directly from historical trajectory data [37]. They avoid complex force modeling and

do not require real-time environmental updates. However, purely data-driven methods face a core limitation: they lack embedded physical structure. Instead of modeling the dynamics governed by well-established physical laws, such models treat orbit prediction as a generic time-series forecasting task. This often leads to physically inconsistent results, such as discontinuous trajectories or abrupt, unrealistic velocity changes, as observed in our experiments (§2.2). Besides these limitations, satellite maneuvers further challenge both physics-based and data-driven approaches. Unlike natural orbital motion, satellite maneuvers are intentional thrust events (e.g., for orbit maintenance or collision avoidance) that introduce abrupt trajectory changes. Their timing, direction, and magnitude are mission-specific and highly variable, making them difficult to model or predict. As shown in our experiments, even a single unmodeled maneuver can cause HPOP's prediction error to increase by more than $10\times$ (§2.2). Meanwhile, maneuvers are driven by mission-specific operational decisions and their timing and impact cannot be inferred from past orbital data, making them difficult for data-driven models to anticipate or correct (§2.2).

To address onboard deployment constraints, physical inconsistency, and maneuver-induced failures, we propose AUTOORBIT, a physics-informed orbit prediction framework for real-time onboard use. AUTOORBIT enforces reliable orbit prediction through a hierarchical decomposition of orbital behavior, constraining global geometry, local dynamics, and discrete events. At the global orbital structure level, we reconstruct a reference orbit by fitting a mean orbital element model to historical GNSS data. This reference captures the periodic and bounded nature of orbital motion, providing a stable trajectory backbone that prevents long-horizon drift. A lightweight neural network then models short-term residual deviations around this reference, enabling robust prediction under noisy GNSS inputs. At the local orbital dynamics level, we model fine-grained spatiotemporal evolution using a Fourier neural operator (FNO), which is well suited for capturing periodic patterns in GNSS sequences. In addition, we impose an acceleration-level physics-informed loss to regularize local dynamics, encouraging smooth and physically plausible velocity and acceleration profiles. At the discrete maneuver events level, we introduce a maneuver-aware correction module that leverages known maneuver timing to apply variational updates to the predicted trajectory. This module explicitly accounts for thrust-induced state changes, ensuring continuity across maneuver boundaries and preventing large deviations caused by unmodeled impulsive forces. We evaluate AUTOORBIT on real traces from three in-orbit satellites. At a 30-minute prediction horizon, AUTOORBIT achieves a position error of 62.71 m, improving accuracy by 60.49% over state-of-the-art baselines, while demonstrating robustness to GNSS noise and good cross-satellite generalization. We further deploy AUTOORBIT on an NVIDIA Jetson Orin Nano [40], showing efficient resource usage with stable power and thermal behavior.

Contributions. This paper makes three main contributions:

- We empirically identify key limitations of physics-based and data-driven methods under realistic conditions.
- We design AUTOORBIT, a physics-informed accurate orbit prediction framework for real-time onboard deployment. It enforces Multi-level modeling ensures physical consistency, and it handles impact of maneuvers effectively.

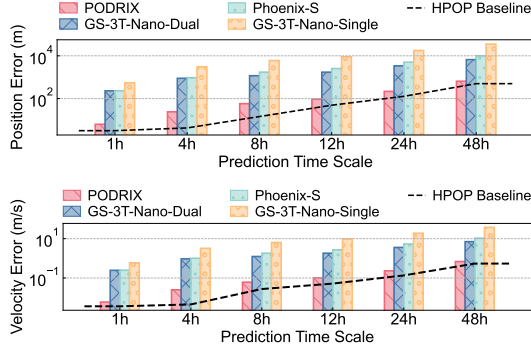


Figure 1: HPOP prediction error under noisy inputs.

- We demonstrate the effectiveness and efficiency of AUTOORBIT through extensive experimental evaluation.

2 Motivation

Orbital motion is complex due to the combined influence of natural perturbations and active maneuvers. Orbit prediction, a challenge in aerospace, is crucial for ground station scheduling and mission planning. Accurate prediction underpins satellite autonomy, enabling onboard task planning [28, 33], formation flying [4], inter-satellite link maintenance [29], and collision avoidance [27].

2.1 Related Works

Current orbit prediction approaches fall into two broad categories: physics-based and data-driven methods [10, 49].

Physics-based methods can be further divided into two subtypes. The first includes simplified analytical or semi-analytical models, which prioritize computational efficiency by modeling only the dominant forces and applying approximations to perturbations. Representative examples include the classical Two-Body model [5] and the widely adopted SGP4 model [20, 53]. The second subtype consists of high-fidelity numerical models [36], such as the HPOP model, which achieve high accuracy by rigorously integrating a comprehensive set of perturbative forces using fine-grained physical parameters and high-order dynamics.

Data-driven methods rely on historical orbital data from individual satellites, using machine learning techniques to forecast future states—typically represented as orbital elements or position-velocity vectors [7, 32, 42]. These models aim to learn spatio-temporal patterns directly from data, bypassing explicit physical modeling.

2.2 Challenges for Orbit Prediction Onboard

Challenge 1: Supporting standalone onboard deployment. Autonomous satellites must predict their orbits onboard in real-time without continuous ground contact or external updates. However, existing physics-based methods face critical deployment barriers. Physics-based models such as HPOP offer high precision by numerically integrating orbital motion with detailed perturbation forces. However, they are highly sensitive to initial-state noise from raw GNSS input data onboard and require continuous updates of

ground-sourced environmental parameters. We evaluate this sensitivity by injecting Gaussian noise matched to the nominal accuracy of four GNSS receivers into the ground-truth orbit states, namely the Precise Orbit Determination (POD) data [26]. The perturbed states are then propagated using HPOP. As shown in Fig. 1, even small initial errors lead to significantly amplified position errors over time [6, 30]. Moreover, accurate predictions require updates of Earth orientation [43], space weather [2, 39], and geomagnetic activity [35], which may be unavailable in orbit due to limited or intermittent communication, hindering onboard deployment.

Challenge 2: Enforcing physical consistency. Purely data-driven orbit prediction models learn patterns directly from historical trajectories without explicitly modeling physical forces. While this enables fast inference and avoids complex force modeling, the lack of physical constraints makes such models unreliable under unseen or out-of-distribution conditions. As shown in Fig. 5, data-driven predictions often exhibit discontinuities and abrupt, physically implausible changes in orbital radius. These artifacts violate fundamental physical laws and differ qualitatively from the smooth error behavior observed in physics-based models. Such physically inconsistent predictions undermine both accuracy and reliability, posing serious risks for safety-critical autonomous tasks such as collision avoidance in congested orbital environments.

Challenge 3: Adapting to scheduled satellite maneuvers. Unlike passive evolution under natural perturbations, many satellites perform maneuvers to adjust their orbits [55], introducing abrupt state changes that challenge prediction. Using Sentinel-1A and Starlink data [25], we highlight three typical scenarios: mission reconfiguration (Fig. 2), routine maintenance [47] (Fig. 3), and collision avoidance [1, 41] (Fig. 4). Physics-based models assume smooth dynamics; maneuvers violate this, causing large errors. As shown in Fig. 6, a maneuver near prediction start increases 24-hour position error to 1348 m—over 10× the baseline. Data-driven models struggle. Fig. 7 shows larger errors on maneuver cases, as their non-physical nature limits predictability from historical data.

3 Physics-Informed Orbit Prediction

3.1 AUTOORBIT Overview

To address the above challenges, we propose AUTOORBIT, a physics-informed orbit prediction framework for real-time onboard applications. The overall workflow is illustrated in Fig. 8. Given onboard GNSS measurements, AUTOORBIT predicts future orbital states over a specified time horizon by hierarchically constraining orbital behavior at three levels: global orbital geometry (Component A), local orbital dynamics (Components B and C), and discrete maneuver events (Component D). Each level introduces targeted inductive biases to ensure long-horizon stability, physically consistent local evolution, and robustness to maneuver-induced disruptions.

Global Orbital Structure Construction (§3.2). To model the global orbit shape, we reconstruct a global reference orbit by fitting a mean orbital element model to historical GNSS data. This reference orbit captures the periodic and bounded nature of satellite motion and prevents long-horizon divergence. Building on this structure, we adopt a residual learning strategy in which a lightweight neural network predicts short-term deviations from the reference orbit, enabling drift-free prediction under GNSS inputs.

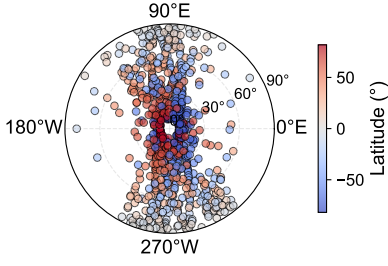


Figure 2: Spatial distribution of Sentinel-1A orbital maneuvers.

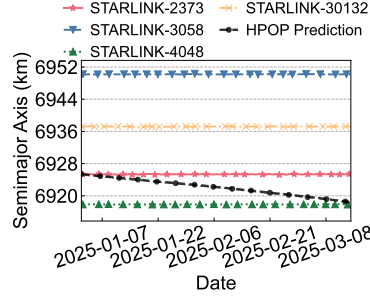


Figure 3: Semi-major axis variations of starlink satellites with time.

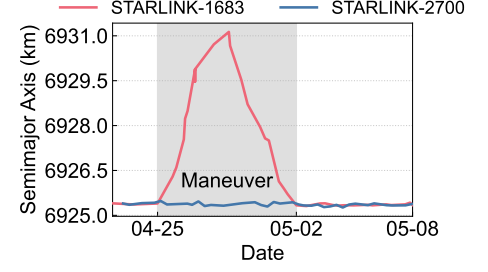


Figure 4: Collision avoidance maneuver.

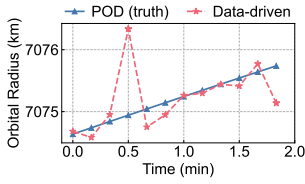


Figure 5: Physical law violation in data-driven method.

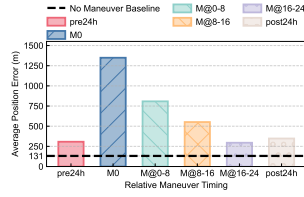
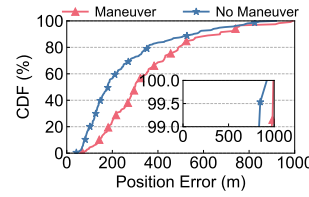
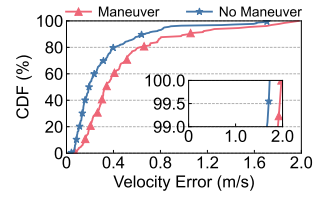


Figure 6: Impact of maneuver on HPOP prediction.



(a) Position Error.



(b) Velocity Error.

Figure 7: Data-driven maneuver prediction errors.

Local Orbital Dynamics Learning (§3.3). To maintain local dynamics consistency, we employ an FNO to learn spatiotemporal patterns from GNSS position sequences, leveraging its ability to capture long-range and periodic dynamics. We further impose a physics-informed loss on second-order orbital acceleration, encouraging smooth and physically plausible velocity and acceleration profiles while mitigating discontinuities in data-driven models.

Real-Time Orbital Maneuvers Correction (§3.4). To adapt to scheduled satellite maneuvers, we incorporate a maneuver-aware correction module that leverages available prior knowledge of maneuver timing and applies a variational update to the predicted trajectory. This correction enforces physical continuity across maneuver events, aligns predictions with post-maneuver orbits, and prevents large deviations induced by unmodeled impulses.

AUTOORBIT achieves high prediction accuracy, strong physical consistency, and lightweight deployment. Instead of directly predicting absolute orbital states with large dynamic ranges, the reference orbit provides a stable global backbone, allowing the neural network to focus on small residual deviations. This design reduces the effective optimization space, improves training stability, and mitigates long-horizon drift. Residual learning and physics constraints work together to improve prediction accuracy, while the FNO-based predictor captures long-range temporal dependencies and periodic patterns in orbital residuals. In addition, maneuver-aware correction explicitly models thrust effects and enhances accuracy during maneuver events. Overall, physical consistency is enforced at three levels: global reference structure, local physics-informed dynamics, and discrete maneuver events level. The self-contained design enables offline training and efficient onboard inference.

3.2 Global Orbital Structure Construction

3.2.1 Reference Orbit Construction. Satellite orbital data is a time series exhibiting temporal periodicity under ideal conditions. However, orbital motion is subject to perturbations including atmospheric drag, solar radiation pressure, and Earth's non-spherical gravitational field, resulting in quasi-periodic characteristics where the orbit does not repeat exactly but exhibits cumulative drift and stochastic perturbations. Nevertheless, the quasi-periodic nature of satellite orbits contains exploitable patterns.

We construct the reference orbit as follows: First, we collect historical orbital observation sequences $\mathbf{s}_{\text{obs}}(t) = (\mathbf{r}_{\text{obs}}(t), \mathbf{v}_{\text{obs}}(t))$ from the onboard GNSS receiver, which are derived from in-orbit measurements without relying on external TLE data or ground-based propagation models. Second, through statistical analysis of historical orbital time-series data, we mine and identify periodic recurrence patterns in the orbit to determine the satellite's orbital recurrence period T_{rec} , which represents the period over which the satellite's ground track repeats relative to Earth's surface. Finally, we align historical orbital data according to this period and apply phase averaging at each orbital phase to extract stable patterns:

$$\mathbf{s}_{\text{ref}}(t_k) = \frac{1}{N} \sum_{i=1}^N \mathbf{s}_{\text{obs}}(t_k + i \cdot T_{\text{rec}}) \quad (1)$$

where N is the number of historical cycles. The resulting reference orbit $\mathbf{s}_{\text{ref}}(t)$ captures the repeatable, periodic component of the satellite's orbital motion — representing the primary dynamical characteristics and long-term evolution trends of the satellite orbit while partially suppressing short-term stochastic perturbations.

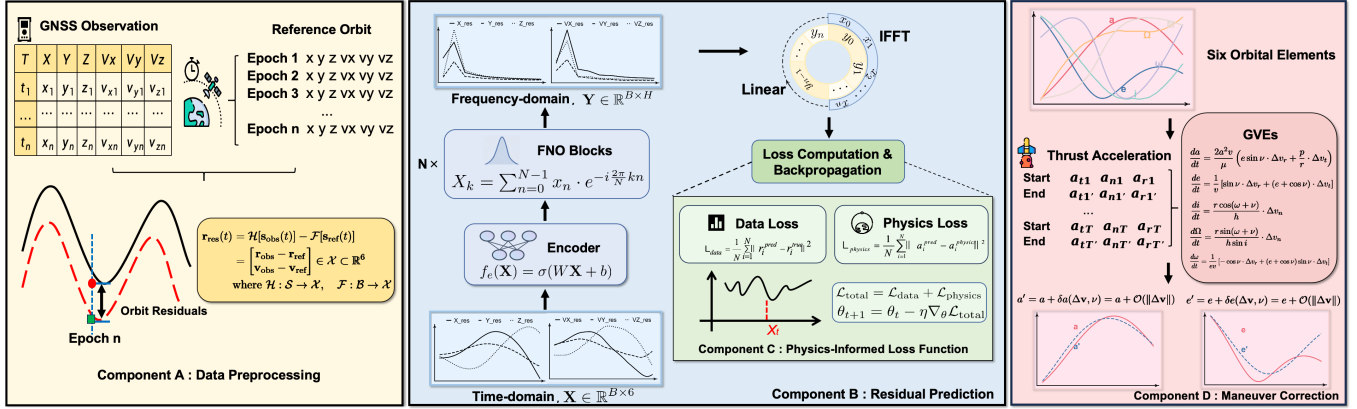


Figure 8: Overview of the AUTOORBIT orbit prediction framework. The system hierarchically constrains orbital behavior at three levels: (A) Global orbit structure, where GNSS observations are preprocessed to extract residuals with respect to a reference orbit; (B–C) Local orbital dynamics, where an FNO-based residual predictor is trained under physics-informed constraints to ensure physically consistent local evolution; (D) Discrete maneuver events, where a maneuver-aware correction module based on GVEs adjusts predictions in response to thrust-induced state changes.

This reference orbit is autonomously computed and stored onboard, providing a stable reference for subsequent residual learning.

3.2.2 Residual Learning Strategy. Predicting absolute orbital states directly is numerically challenging: satellite positions are typically measured in thousands of kilometers and velocities in kilometers per second. Such large dynamic ranges, combined with the complex variability of orbital dynamics and observational noise, lead to unstable gradients, slow convergence, and poor generalization in purely data-driven models. To address this issue, we adopt a residual learning strategy that transforms the prediction task into small-magnitude corrections to the reference orbit.

As illustrated in Fig. 8, the residual is defined as the deviation between the actual observed orbit and the reference orbit:

$$\mathbf{r}_{\text{res}}(t) = \mathcal{H}[\mathbf{s}_{\text{obs}}(t)] - \mathcal{F}[\mathbf{s}_{\text{ref}}(t)] \quad (2)$$

where $\mathcal{H} : \mathcal{S} \rightarrow \mathcal{X}$ and $\mathcal{F} : \mathcal{R} \rightarrow \mathcal{X}$ are mapping functions from the observation space and reference space to the residual space, respectively, and $\mathbf{r}_{\text{res}} \in \mathbb{R}^6$ represents the six-dimensional residual vector of position and velocity. During inference, the predicted orbital state is reconstructed by adding the reference orbit and the predicted residual:

$$\mathbf{s}_{\text{pred}}(t) = \mathbf{s}_{\text{ref}}(t) + \mathbf{r}_{\text{FNO}}(t) \quad (3)$$

where $\mathbf{r}_{\text{FNO}}(t)$ denotes the predicted results by the neural network.

This residual learning framework offers several key advantages: (1) Enhanced physical consistency: leverages the reference orbit’s physical constraints to guide predictions toward analytical and numerical solutions; (2) Historical information mining: extracts temporal periodic patterns and long-term evolution trends from historical data; (3) Improved training stability: small-magnitude residuals reduce optimization dynamic range, improving gradient propagation and accelerating convergence; (4) Enhanced generalization: low-magnitude corrections maintain robustness across variable-horizon prediction tasks.

3.3 Local Orbital Dynamics Learning

3.3.1 Physics-Informed Neural Networks. Physics-Informed Neural Networks (PINNs) [45] integrate machine learning models with physical laws described by partial differential equations (PDEs).

In general, a PINN models the parameterized nonlinear PDE:

$$\psi(t, x) + \mathcal{N}[\psi; \lambda] = 0, \quad x \in \Omega, \quad t \in [0, T] \quad (4)$$

where $\psi(t, x)$ denotes the underlying latent physical field, $\mathcal{N}[\cdot; \lambda]$ is a nonlinear differential operator with parameters λ , Ω is the spatial domain of interest, and T is the final prediction time.

The orbit prediction problem naturally fits the PINN framework: orbital dynamics obey Newtonian mechanics and perturbation theory, providing clear and tractable physical laws, and embedding these priors guides the network toward physically feasible solutions and prevents non-physical outputs. The following subsections introduce the network architecture tailored to orbital dynamics (§3.3.2) and the formulation of the physics-based loss function (§3.3.3).

3.3.2 FNO-Based Prediction Model. We adopt the FNO [31] as the core prediction architecture. FNO was developed for learning solution operators of partial differential equations by directly parameterizing integral kernels in the frequency domain, enabling efficient global convolution operations. For orbit prediction tasks, FNO offers advantages: satellite orbital motion exhibits quasi-periodicity and long-range temporal dependencies, which can be captured through Fourier transforms in the frequency domain.

Input Representation and Frequency Domain Mapping: The model input is a residual sequence $\mathbf{X} \in \mathbb{R}^{B \times 6}$, where B denotes the number of time steps, and each time step contains three-axis position residuals (r_x, r_y, r_z) and three-axis velocity residuals (v_x, v_y, v_z). This time-domain sequence is first mapped to a high-dimensional space via a lifting layer $P : \mathbb{R}^6 \rightarrow \mathbb{R}^{d_v}$, yielding $\mathbf{v}_0 = P(\mathbf{X}) \in \mathbb{R}^{B \times d_v}$.

Frequency Domain Operations in Fourier Layers: As illustrated in Fig. 8, each Fourier layer comprises two parallel pathways. In the frequency domain pathway, the feature $\mathbf{v}_i$ is mapped to the frequency

domain via the Fast Fourier Transform (FFT): $\tilde{\mathbf{v}}_t = \mathcal{F}(\mathbf{v}_t) \in \mathbb{C}^{B \times d_v}$. Since the energy of orbital residuals is primarily concentrated in low-frequency modes (corresponding to the dominant periodic components of the orbit), we retain only the first k_{max} Fourier modes and apply a learnable frequency-domain weight matrix $\mathbf{R}_\phi \in \mathbb{C}^{k_{max} \times d_v \times d_v}$ for linear transformation:

$$\tilde{\mathbf{v}}_{spec} = \mathbf{R}_\phi \cdot \tilde{\mathbf{v}}_t[:, k_{max}, :] \quad (5)$$

This operation efficiently models global temporal dependencies in the frequency domain with computational complexity $\mathcal{O}(B \log B + k_{max} d_v^2)$. The features are then mapped back to the time domain via the Inverse Fast Fourier Transform (IFFT):

$$\mathbf{v}_{spec} = \mathcal{F}^{-1}(\text{ZeroPad}(\tilde{\mathbf{v}}_{spec})) \quad (6)$$

In the time-domain pathway, a local linear transformation $\mathbf{W} : \mathbb{R}^{d_v} \rightarrow \mathbb{R}^{d_v}$ captures local temporal features. The two pathways are fused followed by nonlinear activation:

$$\mathbf{v}_{t+1} = \sigma(\mathbf{v}_{spec} + \mathbf{W}(\mathbf{v}_t)) \quad (7)$$

Autoregressive Prediction Mechanism: Inspired by encoder and decoder architecture, we restructure FNO into an encoder-pooling-decoder framework to accommodate variable-length prediction tasks. The encoder extracts spatiotemporal features through L stacked Fourier layers; the pooling layer compresses the temporal dimension into a latent representation; and the decoder reconstructs the target sequence from the compressed features. Furthermore, we introduce an autoregressive mechanism: the model performs rolling predictions with a fixed time window, outputting residual predictions for the next H steps in each forward pass, which are then appended to the historical sequence for error correction. This design enhances flexibility and stability of long-horizon predictions while preserving FNO's advantages in frequency-domain modeling.

3.3.3 Physics-Informed loss design. To improve physical consistency and generalization of orbit predictions, we incorporate a physics-informed loss that constrains model outputs using known laws of orbital dynamics. The loss is formulated at the *acceleration level*, where perturbation forces influence orbital motion.

We consider four dominant perturbations based on their magnitudes in LEO: Earth's nonspherical gravity (J_2 – J_4 terms), solar and lunar third-body attractions, atmospheric drag, and central gravitation. These parameters are either constant (e.g., J_2 – J_4) or slowly varying (e.g., third-body perturbations and atmospheric density) and do not require real-time updates. Each perturbation is analytically modeled, and the total physical acceleration $\tilde{\mathbf{a}}_{phy}$ is obtained by summing the corresponding perturbation components.

The predicted acceleration $\tilde{\mathbf{a}}_{pred}$ is derived from predicted velocities using a fourth-order finite-difference scheme:

$$\tilde{\mathbf{a}}_{pred} = \frac{-\tilde{\mathbf{v}}_{+2} + 8\tilde{\mathbf{v}}_{+1} - 8\tilde{\mathbf{v}}_{-1} + \tilde{\mathbf{v}}_{-2}}{12\tau}, \quad (8)$$

where $\tilde{\mathbf{v}}_{+k}$ and $\tilde{\mathbf{v}}_{-k}$ denote the predicted velocities at k future and past steps relative to the current time step, and τ is the time interval between consecutive steps, respectively. And then compared with the theoretical physical acceleration $\tilde{\mathbf{a}}_{phy}$ using mean squared error:

$$L_{physics} = \text{MSE}(\tilde{\mathbf{a}}_{pred}, \tilde{\mathbf{a}}_{phy}). \quad (9)$$

By minimizing this loss during training, the model conforms to conservation laws and perturbation dynamics, reducing anomalous predictions under maneuvers or noisy inputs.

3.4 Real-Time Orbital Maneuvers Correction

Satellite maneuvers alter the spacecraft's trajectory by generating acceleration through the propulsion system, which accumulates over a short duration to produce instantaneous velocity increments. However, the velocity jumps introduced by maneuvers cause abrupt changes in orbital elements, violating the dynamics assumption and rendering prediction models based on smooth trajectory evolution ineffective. To address this issue, we propose an embedded real-time maneuver correction algorithm that analytically models the instantaneous and long-term effects of velocity increments on orbital evolution through Gaussian Variational Equations (GVEs).

For velocity increments $\Delta \vec{v} = [\Delta v_r, \Delta v_a, \Delta v_c]^T$ in the RAC (Radial-Along-track-Cross-track) coordinate frame, the complete GVEs expressions are shown in Component D of Fig. 8. GVEs describe the instantaneous rate of change of the six Keplerian orbital elements under small thrust perturbations, where $a, e, i, \Omega, \omega, \nu$ represent the semi-major axis, eccentricity, inclination, right ascension of ascending node, argument of periapsis, and true anomaly, respectively.

For maneuver scenarios of LEO satellites (orbit maintenance, collision avoidance), velocity increments are dominated by radial and along-track components ($\Delta v_r, \Delta v_a \gg \Delta v_c$). GVEs analysis reveals that semi-major axis a , eccentricity e , and argument of periapsis ω exhibit highest sensitivity to radial and along-track impulses, while inclination i and right ascension of ascending node Ω are primarily affected by cross-track impulses. Therefore, in resource-constrained onboard environments, we prioritize the correction of (a, e, ω) to achieve real-time performance while maintaining accuracy.

The algorithm performs orbital correction in two stages. First, at the maneuver epoch t_m , GVEs are applied to compute the instantaneous jumps in orbital elements:

$$\Delta a = \frac{2a^2 v}{\mu} \left(e \sin \nu \cdot \Delta v_r + \frac{p}{r} \cdot \Delta v_a \right) \quad (10)$$

$$\Delta e = \frac{1}{v} [\sin \nu \cdot \Delta v_r + (e + \cos \nu) \cdot \Delta v_a] \quad (11)$$

$$\Delta \omega = \frac{1}{ev} [-\cos \nu \cdot \Delta v_r + (e + \cos \nu) \sin \nu \cdot \Delta v_a] \quad (12)$$

Subsequently, the updated orbital elements $(a', e', \omega') = (a + \Delta a, e + \Delta e, \omega + \Delta \omega)$ are propagated to subsequent epochs via Kepler's equation and transformed back to ECI (Earth-Centered Inertial) Cartesian state vectors. This propagation process implicitly incorporates the maneuver's impact on long-term dynamical characteristics such as orbital period and phase evolution. For instance, changes in the semi-major axis directly modify the orbital period $T = 2\pi\sqrt{a^3/\mu}$, thereby affecting the satellite's spatial position at future epochs; variations in eccentricity and argument of periapsis reshape the orbital geometry. By analytically propagating the updated orbital elements, the algorithm naturally extends the instantaneous effects of the maneuver throughout the entire prediction horizon without requiring additional numerical integration. The computational complexity of a single maneuver correction is $\mathcal{O}(H)$.

Model		POD Based (Ideal)						GNSS Based					
		S1A			S3A			S1A			S3A		
		2022	2023	2024	2022	2023	2024	2022	2023	2024	2022	2023	2024
HPOP	Pos	1.469	1.182	1.197	2.503	1.659	1.292	144.502	180.435	89.931	135.831	90.139	89.249
	Vel	0.002	0.002	0.001	0.004	0.002	0.002	0.216	0.268	0.103	0.170	0.109	0.141
SGP4*	Pos	1615.301	1290.843	1733.594	2187.878	1151.002	1983.994	1615.301	1290.843	1733.594	2187.878	1151.002	1983.994
	Vel	1.668	1.283	1.794	2.178	1.092	1.933	1.668	1.283	1.794	2.178	1.092	1.933
LSTM	Pos	1107.059	808.067	839.440	731.136	628.419	664.879	1334.529	1269.715	1277.883	1170.192	1081.887	1095.888
	Vel	1.289	1.004	1.030	1.737	1.664	1.689	1.453	1.472	1.474	1.994	1.944	1.930
DASR	Pos	2713.931	3900.325	1915.342	2829.784	2762.044	2695.951	2714.080	2718.447	2677.826	2832.954	2700.505	2698.284
	Vel	2.875	3.895	2.644	2.902	2.898	2.829	2.876	2.877	2.868	2.903	2.835	2.830
Informer	Pos	320.793	127.800	159.974	156.506	99.813	121.050	267.275	133.649	153.391	123.735	80.990	89.628
	Vel	0.369	0.162	0.200	0.174	0.113	0.135	0.296	0.164	0.187	0.140	0.098	0.107
AutoOrbit	Pos	68.566	60.605	61.098	59.400	59.740	63.559	67.587	55.342	60.270	58.770	64.234	61.549
	Vel	0.091	0.077	0.079	0.074	0.071	0.086	0.092	0.060	0.083	0.076	0.084	0.079

Table 1: Results of the main experiments. SGP4* uses TLE data as input, so its prediction accuracy varies only across satellites.

Model		Prediction Horizon											
		0.5h		1h		4h		8h		12h		24h	
		S1A	S3A	S1A	S3A	S1A	S3A	S1A	S3A	S1A	S3A	S1A	S3A
LSTM	Pos	872.202	825.319	1024.474	1002.374	1104.405	1059.978	1114.384	1066.433	1102.716	1054.893	1124.950	1060.403
	Vel	0.854	0.799	0.814	0.751	1.059	1.064	1.059	1.066	0.937	0.889	0.975	1.094
Informer	Pos	275.027	268.642	249.851	379.407	455.348	727.599	463.801	742.860	479.315	743.799	471.147	851.203
	Vel	0.320	0.293	0.263	0.431	0.425	0.754	0.448	0.769	0.463	0.767	0.479	0.891
AutoOrbit	Pos	60.270	61.549	123.831	101.943	155.234	152.891	198.567	204.123	242.122	256.706	322.178	343.225
	Vel	0.083	0.079	0.195	0.124	0.238	0.226	0.267	0.263	0.297	0.301	0.335	0.377

Table 2: Experiments on different prediction horizons, covering short-term, medium-term, and long-term prediction scenarios.

(where H is the prediction horizon length), significantly lower than the $O(H \cdot N_{step})$ complexity of numerical integration methods.

4 Experiment

In this section, we evaluate AUTOORBIT through comparison with SOTA baselines and ablation based on real satellite dataset.

4.1 Experimental Setup

Datasets. We evaluate AUTOORBIT on three satellites operating at different orbital altitudes: Sentinel-1A (S1A)[15], Sentinel-3A (S3A)[16] and Sentinel-3B (S3B)[16], using POD and GNSS data collected from 2022 to 2024. To reflect realistic onboard constraints, we conduct experiments under two input settings: (1) high-accuracy POD data and (2) raw GNSS measurements. This dual setup enables comprehensive validation of the model’s applicability across both ideal and practical onboard conditions.

Data Preprocessing. All residuals are normalized using Z-score standardization: $\hat{x} = (x - \mu)/\sigma$, where μ and σ are the mean and standard deviation computed from the training set. These statistics are reused for validation and testing to ensure consistency. We used the data from January to October for training, November for validation, and December for testing.

Baseline Methods. To comprehensively evaluate AUTOORBIT’s performance, we select five representative methods as comparison benchmarks: two physics-base methods: HPOP and SGP4, and

Model		S3B (GNSS Based)		
		2022	2023	2024
HPOP	Pos	82.011	132.165	158.507
	Vel	0.084	0.191	0.238
SGP4	Pos	1796.778	1745.159	2158.550
	Vel	1.784	1.749	2.187
LSTM	Pos	1102.892	1238.869	1367.121
	Vel	1.981	2.066	2.197
DASR	Pos	2734.690	2824.586	2829.062
	Vel	2.845	2.896	2.900
Informer	Pos	97.247	194.618	254.393
	Vel	0.112	0.221	0.284
AutoOrbit	Pos	49.039	68.772	82.121
	Vel	0.071	0.082	0.103

Table 3: Cross-satellite generalization results on Sentinel-3B.

three data-driven methods: LSTM[19], Informer[58], and DASR[22], representing the SOTA of satellite orbit prediction.

Evaluation Metrics. Three-axis position error (m) and velocity error (m/s) are adopted as quantitative evaluation metrics, computed as the position and velocity errors between predicted orbits and ground truth data in the ECI coordinate frame.

Hyperparameter Configuration. After optimization, the final configuration is as follows: batch size of 128, initial learning rate of

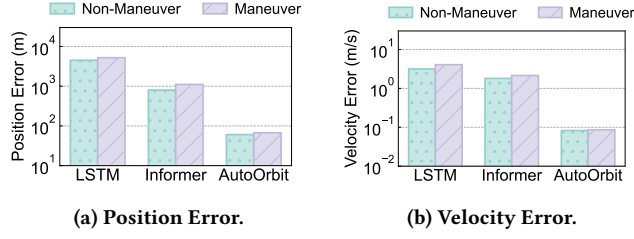


Figure 9: Prediction errors on maneuver days and non-maneuver days in December 2024.

0.001 (decayed to 0.97 times after each epoch), 4 Fourier layers in encoder and decoder (8 layers in total), and training for 10 epochs. **Training and Inference Strategy.** We adopt a seq-to-seq (multi-step) training setup: each input window contains 8640 residual steps (past 24 hours), and the model predicts the next 60 steps (10 minutes) in a single forward pass. The model parameters are optimized with Adam via backpropagation under a joint data loss and physics loss. At inference, we run an autoregressive sliding window: newly predicted temporal steps are appended while the earliest 60 steps are discarded, thereby enabling continuous forecasts.

Platforms. Model training is conducted on a server equipped with an NVIDIA L20 48 GB GPU and deployed on an NVIDIA Jetson Orin Nano computing platform, which offers 128 CUDA cores and 4 GB of RAM, with power consumption capped at 10 W, matching compute envelope of recent flight-validated onboard processors [40].

4.2 Main Experiment Results

Tab. 1 summarizes the prediction performance of three satellites across multiple years under two types of input data.

Results on POD Data. Under ideal POD inputs, AUTOORBIT achieves the second-best accuracy, second only to ground-based HPOP with noise-free inputs. Compared to the best data-driven baseline Informer, AUTOORBIT achieves accuracy improvements of 62.17% and 58.59% for position and velocity prediction, respectively. The results demonstrate that physics-constrained residual learning effectively integrates the prior knowledge of physical models with the expressive power of data-driven methods, significantly improving prediction accuracy while maintaining physical consistency.

Results on GNSS Data. Under realistic GNSS noisy inputs, AUTOORBIT achieves the best overall prediction accuracy. Compared to the best data-driven baseline Informer, it achieves 66.95% and 63.75% improvements in both position and velocity accuracy, respectively, and 49.64% and 53.00% substantial improvements compared to HPOP under the same realistic noisy conditions.

Results across Maneuver and Non-Maneuver Days. To isolate the impact of orbital maneuvers, we partition December 2024 into 8 maneuver days and 23 non-maneuver days and compare per-day prediction errors. As shown in Fig. 9, AUTOORBIT exhibits only 10.4% position and 4.3% velocity degradation on maneuver days, while LSTM and Informer degrade by 16.0% and 38.8% in position and 29.1% and 18.7% in velocity, respectively. The maneuver-aware correction module compensates thrust-induced state changes via GVEs, preserving stability where purely data-driven baselines suffer substantial accuracy losses.

4.3 Different Prediction Horizons Evaluation

In prediction horizons tests ranging from 0.5 hours to 24 hours, AUTOORBIT achieves the best accuracy across all prediction horizons (Tab. 2). Compared to Informer, it achieves an average position accuracy improvement of 79.15% in short cycles (0.5h) and maintains 65.80% improvement in long cycles (24h). The results demonstrate that AUTOORBIT maintains stable prediction superiority across different update cycles.

4.4 Model Generalization

Cross-satellite validation testing on Sentinel-3B (Tab. 3) demonstrates that model trained on Sentinel-3A maintains consistent high prediction accuracy without retraining, validating strong cross-satellite generalization capability of AUTOORBIT within the same constellation. Specifically, AUTOORBIT outperforms the best baseline Informer by 49.6% to 67.7%. This zero-shot transfer capability demonstrates that AUTOORBIT effectively learns mission-agnostic orbital dynamics rather than overfitting to satellite-specific orbits.

4.5 Ablation Study

We perform ablation experiments on 2024 Sentinel-1A data to evaluate key design components. The hidden dimension is fixed at 256, and other settings follow the main configuration (§4.1). We remove the physics-informed loss and maneuver correction modules separately to assess their impact on prediction accuracy.

Ablation Study for Physics Constraints. The physical consistency error quantifies how well predictions conform to orbital dynamics equations: by substituting predicted state vectors into the dynamical PDE equations and computing the equation residuals, with lower values indicating better overall adherence to physical laws. As shown in Fig. 10a and Fig. 10b, after adding the physics-informed loss, the physical consistency error metrics (P95 and P99) improve by 38.7% and 32.9%, respectively, and position and velocity prediction accuracy both significantly improve by 50%.

Ablation Study for Maneuver Correction. As shown in Fig. 11, the maneuver correction module reduces position errors in 61.8% of affected samples and velocity errors in 54.2% of samples, with error increases not exceeding 0.42% in negative cases. The results demonstrate that the maneuver correction module effectively compensates for prediction deviations caused by orbital maneuvers while maintaining overall stability.

4.6 Edge Computing Deployment

We deploy LSTM, Informer, HPOP and AUTOORBIT on an NVIDIA Jetson Nano platform for comparative testing, evaluating onboard deployment feasibility from three dimensions: resource utilization, inference latency, and power-thermal characteristics. The experimental results validate the practical deployment capability of AUTOORBIT in resource-constrained onboard environments. **System Overhead and Inference Latency Analysis.** As shown in Fig. 12, AUTOORBIT achieves inference time of 1.36s for predicting a full 24-hour orbital trajectory, 26.9% faster than LSTM (1.86s) and 75.1% faster than Informer (5.46s), making it the fastest among the three. It exploits hardware acceleration through high GPU utilization, meeting real-time task requirements [8, 11, 21, 34, 38]. Combined with the accuracy results in §4.2, AUTOORBIT achieves the highest

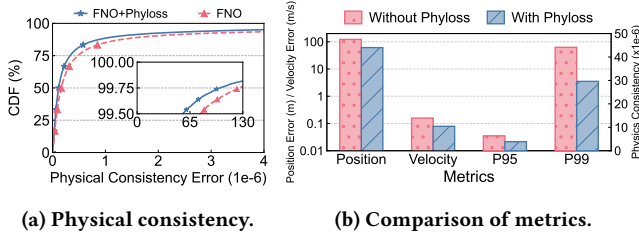


Figure 10: Physical consistency and metrics evaluation.

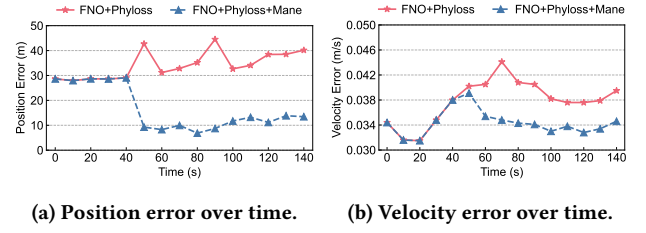


Figure 11: Maneuver correction effects.

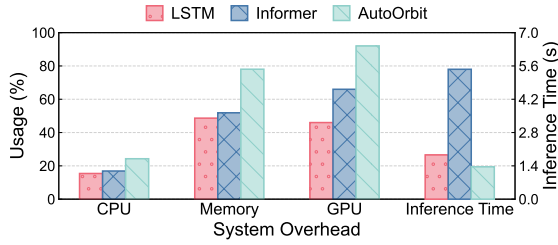


Figure 12: System overhead.

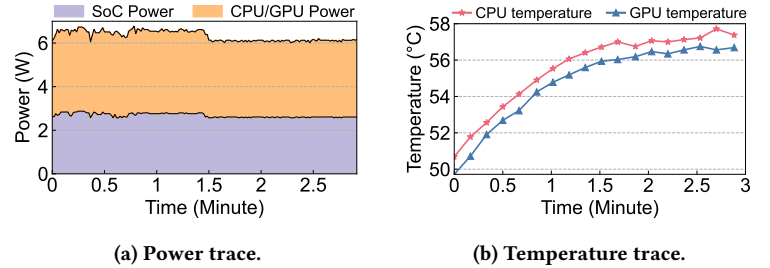


Figure 13: Power and temperature characteristics.

Horizon	0.5h	1h	4h	8h	12h	24h
Inference Time (s)	304.00	411.00	400.00	427.00	461.00	482.00
Power (W)	5.35	5.60	5.68	5.67	5.62	5.74

Table 4: HPOP inference cost on platform. At 24 h, AUTOORBIT completes the same prediction in only 1.36 s at 3.45 W, achieving a 354× speedup and 40% lower power.

prediction accuracy with the shortest inference time, validating effective balance between accuracy and efficiency.

Power and Thermal Characteristics. As shown in Fig. 13, during the process of predicting the orbit for a one-month duration, AUTOORBIT achieves an average power consumption of 6.33W, well below the 10W rated power of Jetson Nano, and a peak temperature below 58°C, far below the 85°C DVFS threshold of flight-validated platforms. The results demonstrate that even under passive thermal control conditions typical of small satellites, COTS hardware running AUTOORBIT maintains excellent thermal stability, ensuring the reliability of long-term onboard operation.

Comparison with HPOP on the edge device. We further benchmark HPOP on the same Jetson Orin Nano. As shown in Tab. 4, HPOP requires 304–482 s across horizons, whereas AUTOORBIT completes a 24 h prediction in only 1.36 s with 40% less power, confirming that AUTOORBIT satisfies both the latency and the energy budgets for real-time onboard deployment.

5 LIMITATIONS AND ETHICAL CONSIDERATIONS

Data used in this study are publicly released by the European Space Agency (ESA) via the Copernicus Data Space for Sentinel satellite

orbits. These datasets are available without access limitations or authorization requirements, and do not involve human participants or sensitive information. This work does not raise ethical concerns.

6 Conclusion

In this paper, we propose AUTOORBIT, a physics-informed autonomous system for onboard real-time orbit prediction. By integrating reference orbit construction, FNO-based residual learning, physics constraints, and maneuver correction, AUTOORBIT significantly improves overall prediction accuracy while maintaining physical consistency. Deployment validation on the in-orbit satellites and resource-constrained edge platforms demonstrates practical deployment readiness as a high-accuracy, efficient solution for onboard orbit prediction. Future work will explore the system’s application to large-scale multi-satellite constellations and mega-constellations, including distributed prediction architectures for constellation-wide coordination and communication. Additionally, extending the proposed framework to other orbit regimes such as medium Earth orbit (MEO) and geostationary orbit (GEO) presents highly promising research directions for broader operational applicability and enhanced mission autonomy.

Acknowledgments

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A Fundamentals Of Orbital Dynamics

A.1 Orbital State Representations

Satellite orbital motion is a nonlinear dynamical system governed primarily by Earth’s central gravitational force and influenced by various small perturbations, such as Earth’s oblateness, atmospheric drag, solar radiation pressure, and third-body effects. Its behavior follows classical Newtonian mechanics. Within the discipline of orbital mechanics, the motion of a satellite can be mathematically described using several equivalent but functionally distinct state representations. This section introduces three widely used formulations, i.e., classical orbital elements, the TLE set, and the state vector representation, each suited for different analytical, operational, and computational purposes.

Classical Orbital Elements Representation. Rooted in Keplerian theory, the classical orbital elements provide a compact, physically interpretable description of a satellite’s orbit. The six parameters [5] (semi-major axis, eccentricity, inclination, right ascension of the ascending node, argument of perigee and mean anomaly) define the orbit’s size, shape, orientation, and the satellite’s position within it [10]. This representation is foundational to orbit design and long-term analysis, as the parameters remain relatively stable over short timescales and evolve predictably under perturbative forces such as atmospheric drag and gravitational harmonics.

TLE Set Representation. The TLE set offers a standardized, low-bandwidth method for disseminating orbital data of Earth-orbiting objects and remains the cornerstone of public satellite tracking infrastructure. Developed by NORAD in the 1970s, a TLE set encodes orbital parameters into two 69-column ASCII lines (including classical orbital elements) into two fixed-length ASCII lines. As an example, the TLE data of Sentinel-3A in December 2024 is shown in Tab. 5. TLE data is published by the 18th Space Defense Squadron of the U.S. Space Force through Space-Track.org [50] and redistributed via CeresTrak.org [24]. For LEO satellites experiencing significant atmospheric drag, such as Sentinel-series satellites, TLEs are updated 2-4 times daily to maintain propagation accuracy. The TLE data is specifically designed for use with simplified perturbations models (SGP, SGP4, SDP4, etc.) [20, 53], allowing any of these SGP models to compute satellite position and velocity at any point in time using TLE data as input. Although generated on the ground, TLEs are updated only a few times per day and can be uplinked during scheduled contacts, making them feasible for onboard use.

State Vector Representation. In contrast to the parameterized formulations above, the state vector representation describes a satellite’s instantaneous kinematic state through its three-dimensional spatial coordinates (X, Y, Z) and corresponding velocity components (V_x, V_y, V_z) at a specific epoch. Compared to orbital elements and TLE formats, State vector representation offers the advantage of direct expression, facilitating coordinate system transformations and physical computations. In operational settings, the state vector representation serves as the standard format for recording and disseminating satellite positions, including real-time outputs from onboard GNSS receivers and post-processed precise orbit determination [48, 59] (POD) data products from Sentinel-series satellites [13]. Consequently, this representation is frequently used as direct input to orbit prediction algorithms, including the HPOP

model, as well as learning-based prediction models, including our system AUTOORBIT.

```
1 41335U 16011A 16047.83464389 -.00000044 00000-0 00000+0 0 9996
2 41335 98.6220 116.2408 0002152 57.4230 96.4527 14.26617832
```

Table 5: Example of TLE set for Sentinel-3A.

Receiver	Type	Pos. (m)	Vel. (m/s)
Phoenix-S ¹	Single	≤ 10	≤ 0.1
OEMV-1G ²	Single	≤ 8.5	≤ 0.1
COSGNSS-Single ³	Single	≤ 5.5	≤ 0.2
Gemini-SR5 ⁴	Single	≤ 3	≤ 0.5
GS-3T-Nano-Single ⁵	Single	≤ 10	≤ 0.2
OEM719 ⁶	Single	≤ 4.5	≤ 0.03
PODRIX ⁷	Dual	≤ 1	≤ 0.002
GS-3T-Nano-Dual ⁸	Single	≤ 10	≤ 0.2
OEM719 ⁹	Dual	≤ 15	≤ 0.1
Orion B16-C ¹⁰	Dual	≤ 5	≤ 0.1
J150 Beidoustar ¹¹	Dual	≤ 10	≤ 0.2
COSGNSS-Dual ¹²	Dual	≤ 3.5	≤ 0.05
NavSBR ¹³	Dual	≤ 7	≤ 0.02

Table 6: Specifications of onboard GNSS receivers.

A.2 Onboard Data Sources

Real-time GNSS Receiver Observations. Onboard GNSS receivers output real-time three-dimensional position and velocity information by receiving signals from navigation satellites such as GPS [23], BeiDou [56], and GLONASS [46]. This data offers the advantage of real-time availability and can be directly used for onboard navigation, positioning, and as input for autonomous orbit prediction. However, due to factors including navigation satellite geometry, receiver performance, and signal propagation errors, GNSS real-time observations contain certain levels of error, as shown in Tab. 6.

POD Data. POD data (As an example, the POD data of Sentinel-3A on December 1, 2024 is shown in Tab. 7) is essentially derived from ground-based post-processing of onboard GNSS observations, obtained through the integration of global ground tracking station observations, precise ephemerides, and geophysical models, using batch least squares, filtering methods, and physical model corrections. Sentinel satellites POD data product [13] achieves a

¹Springer handbook of global navigation satellite systems

²GPS relative navigation for the CanX-4 and CanX-5 formation-flying nanosatellites

³<https://spacef.com/sell/show?id=762>

⁴<https://satsearch.co/products/newspace-systems-ngps-03-422-gps-receiver>

⁵<https://satsearch.co/suppliers/gnssmart>

⁶<https://satsearch.co/products/novatel-oem719-multi-frequency-gnss-receiver>

⁷https://satcatalog.s3.amazonaws.com/components/1216/SatCatalog_-_Beyond_Gravity_-_PODRIX_GNSS_Receiver_-_Datasheet.pdf?lastmod=2023022073318

⁸<https://satsearch.co/suppliers/gnssmart>

⁹<https://satsearch.co/products/novatel-oem719-multi-frequency-gnss-receiver>

¹⁰<https://navspark2.rssing.com/chan-62628515/article156-live.html>

¹¹GNSS receiver for Q-Sat and its analysis of precise orbit determination

¹²<https://spacef.com/sell/show?id=762>

¹³<https://www.moog.com/products/avionics/spacecraft-avionics/gps-receivers/navigation-single-board-receiver-navsbr.html>

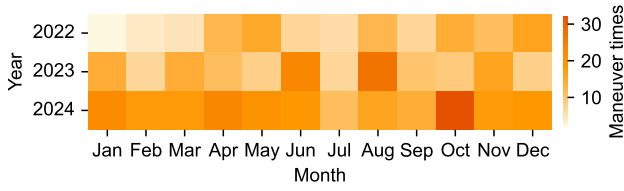


Figure 14: Maneuvers distribution for Sentinel-1A.

Parameter	Value
TAI	2024-12-01 T 22:00:19.000000
UTC	2024-12-01 T 21:59:42.000000
UT1	2024-12-01 T 21:59:42.049786
Absolute Orbit	+45789
X (m)	7126358.480094
Y (m)	-120205.806194
Z (m)	889408.873586
V_X (m/s)	-948.810758
V_Y (m/s)	-1629.210843
V_Z (m/s)	7308.291331
Quality	NOMINAL

Table 7: POD state vector for Sentinel-3A.

3D positional accuracy of 5 cm (1-sigma RMS). Additionally, this data provides quality labels for each orbital data point, e.g., normal maneuver-free points labeled as *NOMINAL* and points affected by maneuvers labeled as *DEGRADED-MANOEUVRE*. Since POD data shares the same origin as GNSS observations but undergoes precise processing, it serves as the ground truth reference for both analysis and orbit prediction accuracy assessment for *AUTOORBIT*.

A.3 Reference Coordinate Systems

Notably, the specific values of a state vector depend on the selected reference frame. Common coordinate systems in orbital prediction include the Earth-Centered Inertial (ECI) [52], Earth-Centered Earth-Fixed (ECEF) [52], and Radial-Along-Track-Cross-Track (RAC) [52] frames. The ECI frame is fixed relative to the stars and is commonly used for orbital dynamics calculations. The ECEF frame rotates with the Earth and aligns with terrestrial geography. It is primarily used for satellite positioning relative to the Earth's surface. The RAC frame is body-fixed to the satellite with three axes point in the radial direction, along-track direction, and cross-track direction. This frame is commonly used for orbital maneuver planning, such as specifying thruster firing directions. In this study, we use POD data provided in the ECEF frame which aligns with the Earth observation mission of Sentinel satellites.

B Satellite Orbital Maneuvers

Beyond passive orbital evolution under natural forces and perturbations, many satellites actively adjust their trajectories through orbital maneuvers[55], which significantly affect the satellite's orbital state. These maneuvers occur across different mission phases and serve various purposes. During the operational phase, they are commonly executed for mission reconfiguration, routine orbit maintenance[47], or collision avoidance[1, 41]. To illustrate their impact, we presents these three representative maneuver scenarios, using publicly available orbital data from the Sentinel-1A and Starlink satellites [25].

Mission Reconfiguration Maneuvers. We analyze the orbital control activity of Sentinel-1A from 2022 to 2024. Sentinel-1A equipped with a synthetic aperture radar payload and must maintain a predefined orbital tube to satisfy imaging geometry and repeat-cycle requirements [47]. As shown in Fig. 14, Sentinel-1A performed 495 maneuvers during this period, averaging about 14 per month—highlighting the high-frequency and routine nature of orbital maintenance. Maneuver timing and location vary with mission planning, orbit maintenance, and collision-avoidance demands. The spatial distribution in Fig. 2 reveals no consistent temporal or geographic patterns, with maneuvers occurring across all orbital phases. This diversity indicates that maneuver behavior is highly irregular and challenging to capture using deterministic modeling approaches.

Orbit Maintenance Maneuvers. For LEO satellites, atmospheric drag[44, 54] introduces significant perturbations due to their low orbital altitude. Without active orbit maintenance, the orbit gradually decays, violating mission design constraints. We analyze the semi-major axis variations of four Starlink satellites at different altitudes over a 60-day period. As shown in Fig. 3, all four satellites maintain nearly constant semi-major axes, indicating active orbit control. Taking Starlink-2373 as an example, we simulate natural orbital decay under atmospheric drag using the HPOP model and compare it with observed data. As illustrated in Fig. 3, orbit maintenance maneuvers compensate for approximately 7 kilometers of decay that would otherwise occur.

Collision Avoidance Maneuvers. When a satellite operator receives a conjunction report[9] from a ground-based observation center indicating that the collision probability with another space object exceeds a predefined threshold within a future time window, a collision avoidance maneuver is typically executed. As an example, Fig. 4 illustrates a conjunction event between Starlink-2700 and Starlink-1683 in May 2022. In response, Starlink-1683 performed an altitude-raising maneuver of approximately 6 kilometers to avoid the predicted collision. After the risk subsided, the satellite gradually returned to its original orbital altitude.